

A Comments and worked examples

Probability theory is a broad and deep area of mathematics and has lots of practical applications. The aim of this course is to provide a solid background in probability theory needed in subsequent courses on subjects including stochastic processes, statistics, machine learning and quantitative finance. The style of the main text is rather concise. This choice is made to (i) convey efficiently the main ideas without distractions, and (ii) show that there is a relatively small set of core concepts and methods but it takes time and effort to master them. Although the notes are logically self-contained, they are not designed to be read alone and you should supplement the notes by reading another textbook. Also, be prepared to read the notes many times as you improve your understanding. For each section we include some comments to discuss further details and references. We also provide some worked examples (but they are not intended to cover all materials covered in the main text).

A.1 Section 1

The first paragraph (about probability theory as a subfield of mathematics) is strongly inspired by the introduction of Feller's book [Fel67]. This book (which has two volumes) is undoubtedly the best introduction to elementary probability theory ever written, and is strongly recommended to all ambitious students. Our discussion of the interpretations of probability is highly simplified. See [DS18] for delightful discussions on various aspects of probability. For subjective/Bayesian probability, our brief discussion in 1.1.2 is motivated by Chapters 1 and 2 of Jayne's book [Jay03]. This book should not be missed if you are interested in understanding some philosophical foundations of probability.

We said that probability is a quantification of uncertainty. What is uncertainty? It appears that there is no single answer. What we can say is that uncertainty can arise in many ways. Here are some examples:

- Coin flip. The process itself can be regarded as deterministic (Newtonian mechanics). However, the outcome is very sensitive to the initial conditions which we cannot control precisely.
- Weather and climate. The processes are much more complicated (fluid dynamics, chaos theory, butterfly effect, etc). Also, at a given point we only have partial information about the current state (e.g. temperatures and wind speeds at isolated locations). Governments, corporations and people may change their behaviours. Long term prediction is difficult.
- Stock market and the economy. We don't even have a clear understanding of the "underlying laws". Also, the available information differs from person to person (this is why insider trading is illegal).
- Pseudo-random numbers. These are routinely used in e.g. computer games and simulations. The numbers are actually not random (they are generated by deterministic algorithms), but behave as if they are "truly random" in the sense that they pass many statistical tests.

- Quantum mechanics. There is *inherent* randomness in nature which cannot be avoided (uncertainty principle).
- etc...

We also note that it may be very difficult to quantify precisely the uncertainty of an event. For example, what is the probability that alien life exists? What is the probability that another pandemic will occur within the next 10 years? While some estimates may be computed based on certain information and assumptions, they must be handled with caution.

A.1.1 Exercises

1. The following R script simulates an experiment where a “fair coin” is tossed “under the same condition” n times.

```
n <- 10000 # number of trials
X <- sample(x = c(0, 1), size = n, replace = TRUE)
# X[1] = result of 1st trial, X[2] = result of 2nd trial, etc.
```

Here n is chosen to be 10000 (feel free to change it). The argument `replace = TRUE` means *sampling with replacement*; this ensures that the trials are performed “independently under the same condition”. By convention, suppose 0 means tail and 1 means head. Write codes to compute the vector of relative frequencies of head: the k -th component is the relative frequency f_k of head in the first k trials. Plot f_k against k . We expect that f_k should approach $\frac{1}{2}$ as k becomes larger and larger.

2. Let A and B be two events such that if A occurs then B must occur as well. We say that A *implies* B . For example, when rolling a dice the event $A = \{\text{number} \geq 4\}$ implies $B = \{\text{number} \geq 2\}$.

Using the frequentist interpretation of probability, justify the property

$$A \text{ implies } B \Rightarrow \Pr(A) \leq \Pr(B).$$

This property, called *monotonicity*, will be proved within the mathematical framework in Proposition 2.9 (ii).

3. Consider the situation depicted in Table 1.1. Let Y be the number of sign changes. For example, the outcome $+++$ has no sign changes (so $Y = 0$ in this case) and the outcome $+- -$ has 1 sign change. Find the value of Y for each outcome and compute $\mathbb{P}(Y = y)$ for all possible values y of Y .